

Adam TAGUIROV

Cybersecurity Researcher & Reverse Engineer

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Profile

Security researcher and software engineer specializing in reverse engineering, malware analysis, and low-level systems.

Built malware analysis platforms based on Xen and custom-instrumented KVM hypervisors, conducted vulnerability research in the Apple ecosystem, and led malware analysis initiatives for Europol and law-enforcement agencies.

Education

École 42 & Paris YNOV Campus, M.Sc. in Information Systems Security Sept 2016 – Sept 2020

- Cybersecurity, operating systems, computer architecture, algorithms, systems and network administration

Professional trainings 2021 – 2023

- Advanced Decryption (CEPOL), Reverse Engineering 101 (Kaspersky), Rust fundamentals (Ambient IT), BADGE Reverse Engineering (ESIEA)

Experience

Founder & Security Researcher, Sigreturn Labs – Paris, France March 2026 – Present

- Founded and operate Sigreturn Labs (sigreturn.com), an independent security research and engineering company.
- Conduct security research, reverse engineering, malware analysis, and software development.
- Design and build security-focused software and infrastructure projects.
- Advise organizations on malware analysis, vulnerability research, and security architecture.

Malware Forensics Specialist, Europol – The Hague, Netherlands May 2025 – March 2026

- Technical lead of EMAS (Europol Malware Analysis Solution), a large-scale malware analysis platform supporting law-enforcement investigations across Europe.
- Designed and developed a malware analysis sandbox based on Xen hypervisor introspection and LibVMI.
- Conducted advanced reverse engineering of malware and ransomware families for international investigations.
- Researched cryptographic and logical weaknesses enabling victim recovery and ransomware decryption capabilities.
- Designed automated malware triage, unpacking, and behavioral analysis workflows.
- Developed tooling and infrastructure supporting operational investigations.

Senior Cybercrime Expert, Judicial Police – Nanterre, France Apr 2023 – May 2025

- Performed reverse engineering of ransomware families used in major cybercrime investigations.
- Researched cryptographic and implementation weaknesses enabling recovery of encrypted victim data.
- Designed and developed decryption tools used in operational investigations.
- Provided malware analysis and incident response expertise across numerous complex cases.

Cybersecurity Researcher, Quarkslab – Paris, France Apr 2022 – Apr 2023

- Conducted vulnerability research on macOS kernel, iOS, WebKit, Chromium-based browsers, and mobile platforms.
- Performed fuzzing, reverse engineering, exploit development, and root-cause analysis.
- Presented public research on JavaScript engine exploitation.
- Completed the ESIEA Reverse Engineering BADGE program.

Cybersecurity Engineer, Orange Cyberdefense – Nanterre, France

Dec 2019 – Apr 2022

- Implemented Windows API hooking, syscall monitoring, process tracking, and memory introspection from the hypervisor layer.
- Developed a second-generation malware analysis platform based on a custom-instrumented KVM stack.
- Modified virtualization components and developed userland tooling enabling memory acquisition, behavioral analysis, and stealth instrumentation of Windows guests.
- Conducted large-scale malware analysis, unpacking, behavioral analysis, and detection engineering.
- Led research efforts and coordinated technical work within a small engineering team.

Cybersecurity Engineer, Digital Security – Paris, France

Sept 2017 – March 2019

- Developed offensive security tooling and binary protection mechanisms for red-team engagements.
- Performed vulnerability research on embedded and IoT systems on ARM architecture.
- Conducted reverse engineering and low-level software analysis.
- Conducted basic hardware reverse engineering and exploitation.

Projects & Community

Root-Me

root-me.org

- Former President and Administrator of Root-Me, one of the largest cybersecurity training platforms in Europe.
- Contributed to platform operations, technical direction, community growth, and security challenge development over an eight-year period.

Polkadot Infrastructure

- Operate production blockchain validator infrastructure securing public decentralized networks.
- Maintain highly available distributed systems, storage, monitoring, and automation infrastructure.
- Participate in governance and technical operations of decentralized blockchain ecosystems.

Publications & Talks

- Speaker - Quarks in the Shell 2023 – JavaScript Engine Exploitation (Paris, France).
- Trainer - International Cooperation with South Africa – Malware Analysis Training (Pretoria, South Africa).
- Trainer - CEPOL session – Lawful Decryption (Budapest, Hungary)
- Speaker - Science Festival at the National Police Museum – Cybercrime Conference (Paris, France)
- Public technical research and publications covering reverse engineering, operating systems, malware analysis, binary instrumentation, software protection, and vulnerability research.
- Portfolio, publications, trainings, and conference materials: taguirov.com/#projects and sigreturn.com/blog

Technical Skills

Programming: C, C++, Python

Security: Reverse Engineering, Malware Analysis, Vulnerability Research, Exploit Development, Fuzzing, Incident Response, Threat Analysis

Systems: Linux Internals, Windows Internals, macOS and iOS, Virtualization, Hypervisors, Xen, KVM

Infrastructure: Distributed Systems, Security Automation, Malware Analysis Platforms, Blockchain Infrastructure

Languages: French (Fluent), English (Fluent, TOEIC 970/990)

Tools: IDA Pro, Ghidra, WinDbg, LLDB, Frida, x64dbg, YARA, Miasm, LibVMI